

Sensors Reference Encyclopedia — Complete Edition

The Full Superset: Course-Notes Material and All Topics from Lectures 13, 14, and 16
 Derivations, worked examples, and Python demonstrations for every concept in Sets 1 and 2

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1 Introduction and How to Use This Document

This is the complete, self-contained reference for the sensors unit: the union of the Set 1 encyclopedia (course-notes material that maps onto the lectures) and the Set 2 encyclopedia (all lecture topics, including those absent from the notes). Nothing from either set has been omitted; where the two sets treated the same topic at different depths, the fuller treatment is used. Each section is self-contained, states its governing equations explicitly, and works numerical examples in full, so you can jump straight to whatever you are stuck on.

Structure: **Part I** — measurement fundamentals (sensors and signal chains, precision/accuracy/resolution, the observer effect, calibration). **Part II** — resistive sensing and readout (thermistors, the voltage divider and the complete ADC chain, the Wheatstone bridge, RTDs, sensitivity, noise). **Part III** — chemical and light-based sensors (smoke, gas, humidity; LDR / photodiode / phototransistor orientation). **Part IV** — optical biosensing and pulse oximetry (Beer–Lambert, PPG, the MAX30102 AFE). **Part V** — op-amp interface electronics. **Part VI** — signal processing (Fourier analysis, filters, the PPG pipeline). Appendix **A** lists corrections to the source notes, which are applied silently throughout the body; Appendix **B** is a consolidated formula sheet.

Conventions. Temperatures in formulas are absolute: $T [\text{K}] = T [^{\circ}\text{C}] + 273.15$. Worked examples default to the lecture demo hardware ($V_{\text{in}} = V_{\text{ref}} = 3.3 \text{ V}$, $R_1 = 10 \text{ k}\Omega$ fixed resistor, $10 \text{ k}\Omega/\beta=3950$ NTC thermistor, 10-bit ADC on an Arduino Nano 33 BLE); the notes' $5 \text{ V}/4.7 \text{ k}\Omega$ configuration appears where the original notes examples are corrected.

PART I

Measurement Fundamentals

2 Sensor Fundamentals

2.1 What a sensor does

A **sensor** converts a physical stimulus — temperature, light, pressure, motion, magnetic field, chemical concentration — into an **electrical output**. The stimulus has physical qualities (intensity, frequency, direction); the electrical output is a voltage, current, resistance, or a stream of digital data. Between the raw transduction and the quantity you actually use sits **signal conditioning**: amplification, filtering, isolation, rectification, and analog-to-digital (AD) conversion.

Two output styles dominate:

- **Analog output:** a continuous voltage over some *range*, proportional (not necessarily linearly!) to the measured property.
- **Digital output:** the measurement is quantized to a fixed number of bits (*resolution*) and delivered as data, often over a serial bus such as I²C or SPI.

2.2 How sensors enter circuits

The notes and Lecture 13 identify two canonical patterns:

1. **Variable passive element.** The sensor behaves as a resistor, capacitor, or inductor whose value tracks the stimulus (thermistor, photoresistor, capacitive humidity element). Since a bare resistance is not directly measurable by a microcontroller, the element is embedded in a network — most often a *voltage divider* (Section 7) — that converts the value change into a voltage change.
2. **Simple switch.** The sensor changes state (open/closed) when the stimulus crosses a threshold (reed switch, tilt switch, tactile switch). The readout is a digital input pin with a pull-up or pull-down resistor.

2.3 Data gathering with a microcontroller

Microcontrollers gather sensor data. Analog signals must first pass through an analog-to-digital converter (ADC). The strongly recommended architecture is a microcontroller with a *built-in* ADC (the Arduino Nano 33 BLE used in the lecture demo has one); bolting a discrete external ADC onto a micro without one works but complicates the design and should be avoided when you can simply choose a better microcontroller. Some sensors integrate their own ADC and calibration and present a purely digital interface — these connect directly to a data port and sidestep the analog chain entirely.

Key Idea 2.1

The full measurement chain you must be able to analyze quantitatively is

$$\begin{array}{ccccc} \text{stimulus} & \xrightarrow{\text{transduction}} & R(T) & \xrightarrow{\text{divider / bridge / amplifier}} & V_{\text{out}} \\ & \xrightarrow{\text{ADC}} & \text{counts} & \xrightarrow{\text{math / LUT}} & \text{engineering units.} \end{array}$$

Section 7.2 works this chain end to end in both directions.

3 Precision, Accuracy, and Resolution

3.1 Definitions

- **Accuracy** is closeness to the true value: how far the reported measurement deviates from an accepted standard.
- **Precision** is the fineness and repeatability of the reported value — loosely, “how many digits.” A bathroom scale that reads 85.7 kg when you weigh 92.1 kg is precise (three significant figures, repeatable) but inaccurate (about 7% low). Precision without accuracy is fine for *relative* measurements on the same instrument, but misleading as an absolute measurement.
- **Resolution** is the smallest distinguishable change. For an analog sensor the output is continuous; once digitized, the reading is *quantized* into 2^N levels for an N -bit converter: 8 bits = 1 part in 256, 10 bits = 1 part in 1024, 12 bits = 1 part in 4096.

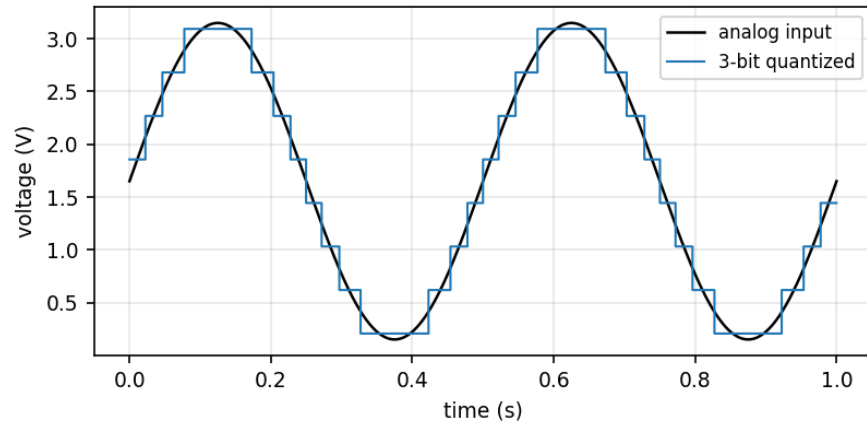


Figure 1: Quantization: a continuous analog input (black) and its 3-bit digitized version (blue). Each step is one least-significant bit (LSB); information between steps is lost.

3.2 Quantization step (LSB)

For an ADC with reference (full-scale) voltage V_{ref} and N bits, one least-significant bit corresponds to

$$\Delta V = \frac{V_{\text{ref}}}{2^N}, \quad \text{counts} = \left\lfloor \frac{V_{\text{out}}}{V_{\text{ref}}} 2^N \right\rfloor. \quad (1)$$

The worst-case quantization error is $\pm \frac{1}{2}$ LSB.

Worked Example 3.1 (LSB of the lecture demo ADC)

The Nano 33 BLE default configuration is a 10-bit ADC with a 3.3 V reference. From Eq. (1),

$$\Delta V = \frac{3.3}{2^{10}} = \frac{3.3}{1024} = 3.22 \text{ mV}.$$

Reconfigured to 12 bits, $\Delta V = 3.3/4096 = 0.806 \text{ mV}$ — four times finer. Note that finer resolution does *not* imply better accuracy: a mis-calibrated 12-bit system is precisely wrong.

Worked Example 3.2 (Resolution in engineering units)

Resolution in volts only matters through the sensor's slope. Near 25 °C the demo divider output changes by $|dV_{\text{out}}/dT| \approx 36.7 \text{ mV}/^\circ\text{C}$ (derived in Worked Example 7.4). Then one 10-bit LSB corresponds to a temperature step of

$$\Delta T \approx \frac{3.22 \text{ mV}}{36.7 \text{ mV}/^\circ\text{C}} \approx 0.088 ^\circ\text{C}.$$

This is the smallest temperature change the digitized system can resolve near room temperature, independent of how accurate it is.

4 The Observer Effect

The **observer effect** states that the act of measuring a property changes it. The classic mechanical example is a tire-pressure gauge: it releases a little air, so the pressure you read is slightly below the pressure that existed before you measured. For most electronic sensors the perturbation is negligible, but it is worth a quick estimate whenever accuracy matters.

The electrical version of this effect for a thermistor is **self-heating**: driving current through the thermistor to measure it dissipates power, which warms the thermistor above the temperature you are trying to measure.

Worked Example 4.1 (Thermistor self-heating in the demo divider)

In the 3.3 V divider with $R_1 = R = 10 \text{ k}\Omega$ at 25°C , the thermistor sees half the supply, so it dissipates

$$P = \frac{(V_{\text{in}}/2)^2}{R} = \frac{(1.65)^2}{10^4} = 272 \mu\text{W}.$$

A small glass-bead thermistor in still air has a dissipation constant of roughly $\delta \approx 1.5 \text{ mW}/^\circ\text{C}$, so the self-heating error is

$$\Delta T = \frac{P}{\delta} \approx \frac{0.272 \text{ mW}}{1.5 \text{ mW}/^\circ\text{C}} \approx 0.18^\circ\text{C}.$$

Acceptable for a classroom demo; not acceptable for a $\pm 0.1^\circ\text{C}$ clinical thermometer, which would use a larger divider resistance or pulsed excitation.

5 Calibration

5.1 The principle

Calibration compares the sensor of unknown accuracy against a **standard of known accuracy**: take a set of readings while the sensor is exposed to known reference values, tabulate the deviation, and compensate. For a temperature sensor the standard might be boiling distilled water at 100°C (at sea-level pressure), or better, a precision calibration oven — which must itself have been calibrated against a still more accurate standard (traceability).

Economics decide the strategy: for a mass-produced consumer product, per-unit calibration is prohibitively expensive, so designs either tolerate the spread between individual sensors or use parts that are **factory-calibrated**, where the manufacturer writes correction constants into on-chip ROM during production (laser trimming and calibration of IC humidity sensors is a common example). Specialized, high-value equipment can justify individual calibration.

5.2 Lookup tables and interpolation

Because the sensor almost always feeds a microcontroller, the standard software mechanism is a **lookup table** (LUT): a table of raw ADC values (left column, e.g., a 10-bit reading between 0 and 1023, perhaps in coarse increments) against the corresponding *accurate* value in engineering units (right column). Between table entries, use **linear interpolation** — assume the short segment of the curve between two entries is a straight line:

$$y(x) = y_1 + (y_2 - y_1) \frac{x - x_1}{x_2 - x_1}, \quad x_1 \leq x \leq x_2. \quad (2)$$

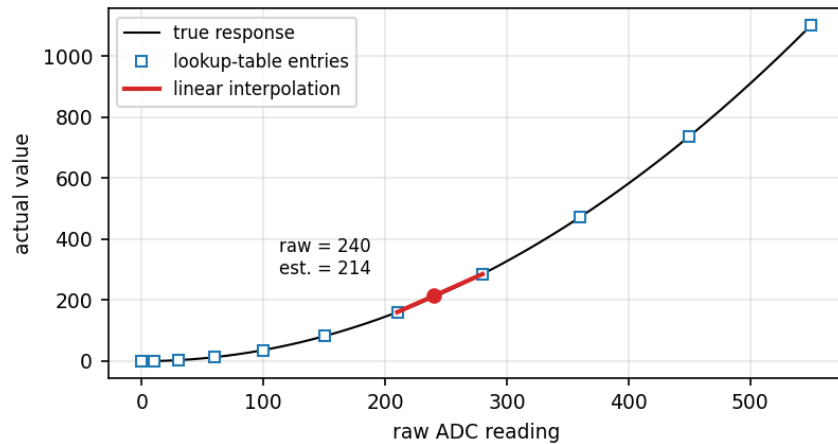


Figure 2: A nonlinear sensor response (black), its lookup-table entries (squares), and linear interpolation between the entries bracketing a raw reading of 240.

Worked Example 5.1 (Interpolating a lookup table)

A calibration LUT contains the entries $(x_1, y_1) = (210, 160.5)$ and $(x_2, y_2) = (280, 285.4)$. The ADC reports $x = 240$. By Eq. (2),

$$y = 160.5 + (285.4 - 160.5) \frac{240 - 210}{280 - 210} = 160.5 + 124.9 \times \frac{30}{70} = 160.5 + 53.5 = 214.0.$$

The interpolation error is the gap between the true (curved) response and the chord; it shrinks quadratically as table entries are spaced more closely.

Listing 1: Building and querying a lookup table in Python.

```
import numpy as np

raw = np.array([ 0, 10, 30, 60, 100, 150, 210, 280, 360, 450, 550])
actual = 0.00364 * raw**2 # "true" calibrated values (from the standard)

def lut_lookup(x):
    """Linear interpolation into the calibration table."""
    return np.interp(x, raw, actual)

print(lut_lookup(240)) # -> 214.0 (chord approximation)
```

```
print(0.00364 * 240**2)           # -> 209.7 (true value; error ~2%)
```

5.3 Extracting model constants from calibration data

For thermistors (next section) there is a smarter alternative to a raw LUT: fit the physical model. Taking the logarithm of the β model $R = R_0 e^{\beta(1/T - 1/T_0)}$ gives

$$\ln R = \underbrace{\beta}_{\text{slope}} \cdot \frac{1}{T} + \underbrace{\left(\ln R_0 - \frac{\beta}{T_0} \right)}_{\text{intercept}}, \quad (3)$$

i.e., $\ln R$ is *linear* in $1/T$. Plot the calibration points as $\ln R$ versus $1/T$ and the slope of the least-squares line is β directly. This is exactly the analysis annotated in Lecture 13's printed-thermistor calibration slide.

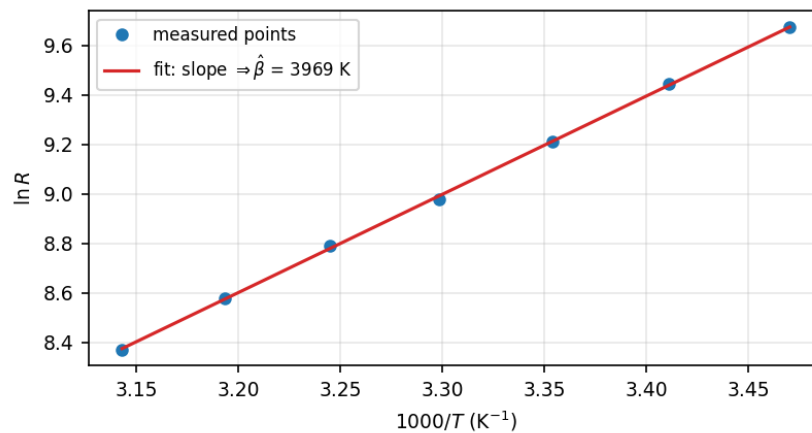


Figure 3: Calibration by model fitting: seven measured (T, R) points (1% resistance noise) plotted as $\ln R$ vs $1000/T$; the fitted slope recovers $\hat{\beta} = 3969$ K against a true value of 3950 K.

Listing 2: Fitting β from measured (T, R) pairs.

```
import numpy as np

T_C = np.array([15, 20, 25, 30, 35, 40, 45.0])           # known oven temperatures
R    = np.array([15837, 12565, 9978, 7980, 6482, 5259, 4350.0]) # measured ohms

x = 1.0 / (T_C + 273.15)           # 1/T in 1/K
y = np.log(R)
slope, intercept = np.polyfit(x, y, 1)
beta = slope
R0    = np.exp(intercept + beta / 298.15) # back out R0 at T0 = 25 C
print(f"beta = {beta:.0f} K, R0 = {R0:.0f} ohm")
# -> beta = 3964 K, R0 = 9981 ohm (true part: 3950 K, 10 kohm; 0.8% noise)
```

PART II

Resistive Sensing and Readout (Lecture 13)

6 Thermistors

6.1 NTC and PTC devices

A **thermistor** (“thermal” + “resistor”) is a resistor whose resistance changes markedly with temperature. Two flavors exist:

- **NTC** (negative temperature coefficient): resistance *decreases* as temperature increases. This is the common type and the one used throughout the course.
- **PTC** (positive temperature coefficient): resistance *increases* with temperature; often used as resettable fuses and heater regulators.

The R–T relationship is strongly *nonlinear*: even over 0–100 °C, a linear approximation introduces large errors (Figure 4); a 10 k Ω / β =3950 part swings from $\sim$ 33 k Ω at 0 °C to $\sim$ 0.97 k Ω at 100 °C.

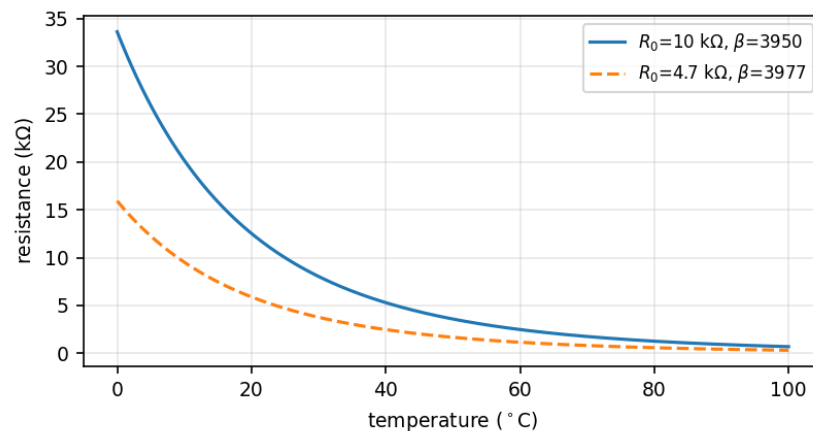


Figure 4: NTC resistance vs. temperature for the lecture demo part (solid) and the notes’ example part (dashed), computed from the β model.

6.2 The Steinhart–Hart equation

The most accurate common model is the third-order Steinhart–Hart equation, valid for both NTC and PTC devices:

$$\frac{1}{T} = A + B \ln R + C (\ln R)^3, \quad (4)$$

with T in kelvin, R the thermistor resistance, and A, B, C constants supplied by the manufacturer (or fitted from three calibration points). Typical accuracy is a few millikelvin over a 100 °C span.

6.3 The β model

The more common single-parameter model assumes constants T_0 (usually $25^\circ\text{C} = 298.15\text{ K}$) and R_0 (the resistance at T_0):

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{\beta} \ln\left(\frac{R}{R_0}\right). \quad (5)$$

Solving for R : multiply through by β , exponentiate, and rearrange to obtain the resistance form

$$R = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}. \quad (6)$$

Datasheets supply β (e.g., 3950 K for the MF58 part used in the lecture demo, 3977 K in the notes), T_0 , R_0 , a temperature range, and an accuracy. The β model is the Steinhart–Hart equation with $C = 0$, $B = 1/\beta$, and $A = 1/T_0 - (\ln R_0)/\beta$.

Worked Example 6.1 (Resistance at body temperature)

Demo part: $R_0 = 10\text{ k}\Omega$, $\beta = 3950\text{ K}$, $T_0 = 298.15\text{ K}$. At $T = 37^\circ\text{C} = 310.15\text{ K}$, Eq. (6) gives

$$R = 10^4 \exp\left[3950\left(\frac{1}{310.15} - \frac{1}{298.15}\right)\right] = 10^4 \exp(3950 \times (-1.2977 \times 10^{-4})) = 10^4 e^{-0.5126} = 5.99\text{ k}\Omega.$$

Worked Example 6.2 (Temperature from a measured resistance)

The same part measures $R = 8.00\text{ k}\Omega$. From Eq. (5),

$$\frac{1}{T} = \frac{1}{298.15} + \frac{1}{3950} \ln \frac{8000}{10000} = 3.35402 \times 10^{-3} + \frac{-0.22314}{3950} = 3.29752 \times 10^{-3}\text{ K}^{-1},$$

so $T = 303.26\text{ K} = 30.1^\circ\text{C}$. Sanity check: $R < R_0$ for an NTC means $T > 25^\circ\text{C}$. ✓

7 The Voltage-Divider Readout

7.1 Derivation

To read a thermistor with a microcontroller ADC we need a voltage. Place the NTC thermistor R at the *top* of a potential divider and a fixed resistor R_1 (conventionally chosen equal to R_0) at the bottom, with V_{out} taken across R_1 :

$$V_{\text{out}} = V_{\text{in}} \frac{R_1}{R_1 + R}. \quad (7)$$

As temperature rises, the NTC resistance R falls, the denominator shrinks, and V_{out} *rises* — the output is a monotonically increasing (though nonlinear) function of temperature. Substituting the β model, Eq. (6):

$$V_{\text{out}} = V_{\text{in}} \frac{R_1}{R_1 + R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}}, \quad \text{and if } R_1 = R_0: \quad V_{\text{out}} = \frac{V_{\text{in}}}{1 + e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}}. \quad (8)$$

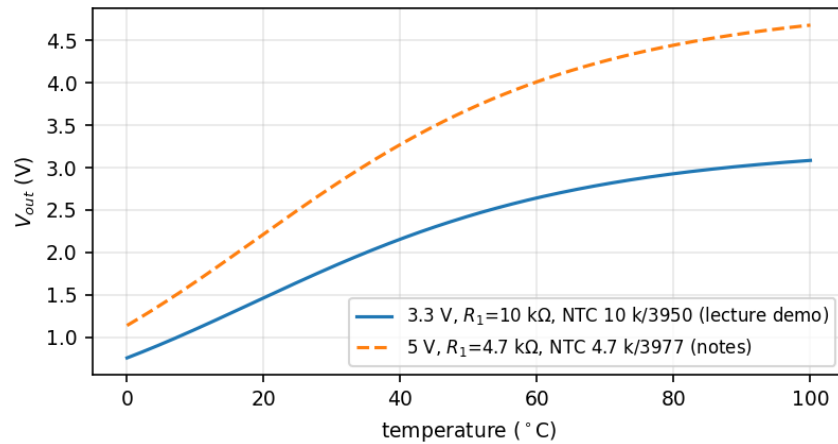


Figure 5: Divider output vs. temperature for the lecture demo configuration (solid) and the corrected notes configuration (dashed). Choosing $R_1 = R_0$ centers the steepest, most sensitive part of the curve at $T_0 = 25^\circ\text{C}$, where $V_{\text{out}} = V_{\text{in}}/2$.

Worked Example 7.1 (Sanity check at T_0 — corrected notes Example 28.2)

At $T = T_0$ the exponent in Eq. (8) is zero, so with $R_1 = R_0$,

$$V_{\text{out}} = \frac{V_{\text{in}}}{1 + e^0} = \frac{V_{\text{in}}}{2}.$$

By definition the thermistor's resistance at 25°C equals R_0 , so the divider splits the supply exactly in half: 2.5 V on a 5 V supply, 1.65 V on 3.3 V. Any formula you derive must reproduce this.

Worked Example 7.2 (Corrected notes Example 28.3: V_{out} at 0°C)

Notes configuration: $V_{\text{in}} = 5\text{ V}$, $R_1 = R_0 = 4.7\text{ k}\Omega$, $\beta = 3977\text{ K}$. (The original solution wrote $25 + 237$ in the exponent — a typo for 273 — and used 273 rather than 273.15; both are fixed here.) At $T = 0^\circ\text{C} = 273.15\text{ K}$:

$$\beta \left(\frac{1}{273.15} - \frac{1}{298.15} \right) = 3977 \times 3.0698 \times 10^{-4} = 1.2208, \quad e^{1.2208} = 3.390.$$

$$V_{\text{out}} = \frac{5}{1 + 3.390} = \frac{5}{4.390} = 1.14\text{ V}.$$

(The notes report 1.15 V from the inconsistent constant; the correct value is 1.14 V.)

Worked Example 7.3 (Lecture demo at body temperature)

Demo configuration: $V_{\text{in}} = 3.3\text{ V}$, $R_1 = 10\text{ k}\Omega$. Using $R(37^\circ\text{C}) = 5.99\text{ k}\Omega$ from Worked Example 6.1:

$$V_{\text{out}} = 3.3 \times \frac{10}{10 + 5.99} = 3.3 \times 0.6254 = 2.06\text{ V}.$$

A finger touch raising the bead from room temperature toward skin temperature moves V_{out} from ~ 1.65 V toward ~ 2 V — exactly the upward step seen in the lecture’s serial-plotter traces.

Worked Example 7.4 (Sensitivity of the divider)

Differentiating Eq. (8) with $R_1 = R_0$ and using the chain rule with $dR/dT = -\beta R/T^2$:

$$\frac{dV_{\text{out}}}{dT} = V_{\text{in}} \frac{R_1 \beta R}{(R_1 + R)^2 T^2}.$$

At $T = T_0$ with $R = R_1$ this collapses to the compact form

$$\left. \frac{dV_{\text{out}}}{dT} \right|_{T_0} = \frac{V_{\text{in}} \beta}{4 T_0^2} = \frac{3.3 \times 3950}{4 \times (298.15)^2} = \frac{13035}{355574} = 36.7 \text{ mV}/^\circ\text{C}.$$

Numerical check: $V_{\text{out}}(26^\circ\text{C}) - V_{\text{out}}(25^\circ\text{C}) = 1.6865 - 1.6500 = 36.5 \text{ mV}$. ✓ This is the slope used in Section 3 to convert ADC resolution into temperature resolution.

7.2 The complete chain: ADC counts to temperature

Putting Sections 3–7 together, the microcontroller inverts the chain in three steps:

$$\text{counts} \rightarrow V_{\text{out}} : \quad V_{\text{out}} = \frac{\text{counts}}{2^N} V_{\text{ref}}, \quad (9)$$

$$V_{\text{out}} \rightarrow R : \quad R = R_1 \left(\frac{V_{\text{in}}}{V_{\text{out}}} - 1 \right) \quad (\text{inverting Eq. (7)}), \quad (10)$$

$$R \rightarrow T : \quad \frac{1}{T} = \frac{1}{T_0} + \frac{1}{\beta} \ln \frac{R}{R_0}. \quad (11)$$

(When the divider runs from the same rail as the ADC reference, $V_{\text{ref}} = V_{\text{in}}$ and the ratio $V_{\text{out}}/V_{\text{in}}$ is simply $\text{counts}/2^N$ — supply drift then cancels, a property called *ratimetric* operation.)

Worked Example 7.5 (Full inversion of one ADC sample)

The demo board reports a 10-bit count of 640.

$$V_{\text{out}} = \frac{640}{1024} \times 3.3 = 2.0625 \text{ V}; \quad R = 10^4 \left(\frac{3.3}{2.0625} - 1 \right) = 10^4 (1.600 - 1) = 6.00 \text{ k}\Omega;$$

$$\frac{1}{T} = \frac{1}{298.15} + \frac{\ln 0.600}{3950} = 3.35402 \times 10^{-3} - 1.29325 \times 10^{-4} = 3.22469 \times 10^{-3} \text{ K}^{-1},$$

so $T = 310.11 \text{ K} = 37.0^\circ\text{C}$. A count of 640 on this hardware means body temperature.

Listing 3: Counts-to-temperature conversion, vectorized.

```
import numpy as np

VIN, VREF, NBITS = 3.3, 3.3, 10
R1, R0, BETA, T0 = 10e3, 10e3, 3950.0, 298.15
```

```
def counts_to_tempC(counts):
    v = counts / 2**NBITS * VREF
    R = R1 * (VIN / v - 1.0)          # invert the divider
    invT = 1.0/T0 + np.log(R / R0) / BETA  # invert the beta model
    return 1.0/invT - 273.15

print(counts_to_tempC(np.array([512, 640, 700])))
# -> [25.00, 36.96, 43.41] degrees C
```

8 The Wheatstone Bridge

The divider's weakness: the interesting signal rides on a large offset ($V_{in}/2$), so a small resistance change produces a small change on a big pedestal. The **Wheatstone bridge** is two dividers side by side, with the output taken *between* their midpoints, so the pedestal subtracts off and $V_o = 0$ when the bridge is balanced.

With supply V_s , arms R_1, R_2 (right divider) and R_3, R_4 (left divider), and V_o the difference of the two midpoint voltages:

$$V_o = V_s \left(\frac{R_1}{R_1 + R_2} - \frac{R_3}{R_3 + R_4} \right) = V_s \frac{R_1 R_4 - R_2 R_3}{(R_1 + R_2)(R_3 + R_4)}. \quad (12)$$

$V_o = 0$ iff $R_1 R_4 = R_2 R_3$ (the balance condition). Now let one arm be a sensor: $R_1 \rightarrow R_1 + \delta R$. Substituting into Eq. (12),

$$V_o = V_s \frac{(R_1 + \delta R)R_4 - R_2 R_3}{(R_1 + \delta R + R_2)(R_3 + R_4)}. \quad (13)$$

Starting from a balanced, equal-arm bridge ($R_1 = R_2 = R_3 = R_4 = R$) this simplifies to

$$V_o = V_s \frac{\delta R}{4R + 2\delta R} \approx V_s \frac{\delta R}{4R} \quad (\delta R \ll R). \quad (14)$$

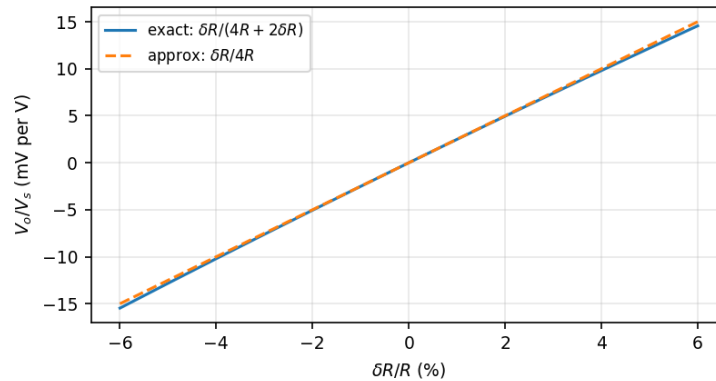


Figure 6: Exact bridge output vs. the small-signal approximation; they are indistinguishable for $|\delta R/R|$ up to a few percent.

Worked Example 8.1 (Strain-gauge bridge)

A $350\ \Omega$ strain gauge in an equal-arm bridge on $V_s = 3.3\text{ V}$ changes by $\delta R = 0.7\ \Omega$ (0.2%) at full load. Approximation: $V_o \approx 3.3 \times 0.7 / (4 \times 350) = 1.650\text{ mV}$. Exact: $V_o = 3.3 \times 0.7 / (1400 + 1.4) = 1.648\text{ mV}$ — the approximation errs by only 0.1%. The millivolt-level output is why bridges are followed by amplifiers (Part V).

Worked Example 8.2 (Why the bridge beats the divider)

For the same 0.2% resistance change, a plain divider output moves from $V_s/2$ by $\approx V_s \delta R / 4R = 1.65\text{ mV}$ — the same *signal* — but sitting on a 1.65 V offset. Amplifying the divider output $\times 1000$ to make the signal visible would also amplify the offset to 1650 V (impossible); the bridge removes the offset first, so the 1.65 mV can be amplified directly.

9 Resistance Temperature Detectors (RTDs)

An RTD is a precision platinum resistor; the ‘pt100’ has $R_0 = 100\ \Omega$ at 0°C . Its resistance follows the Callendar–Van Dusen polynomials:

$$R_T = R_0 [1 + AT + BT^2 + (T - 100)CT^3] \quad -200^\circ\text{C} < T < 0^\circ\text{C}, \quad (15)$$

$$R_T = R_0 [1 + AT + BT^2] \quad 0^\circ\text{C} < T < 661^\circ\text{C}, \quad (16)$$

with standard coefficients $A = 3.9083 \times 10^{-3}\text{ K}^{-1}$, $B = -5.775 \times 10^{-7}\text{ K}^{-2}$, $C = -4.183 \times 10^{-12}\text{ K}^{-4}$ (here T is in $^\circ\text{C}$). Higher-order coefficients can be found by calibrating the individual resistor. Compared to a thermistor, the RTD is far more linear and stable but much less sensitive ($\sim 0.39\%/^\circ\text{C}$ vs. $\sim 4\%/^\circ\text{C}$), which is why precise RTD measurement uses a Wheatstone bridge front end.

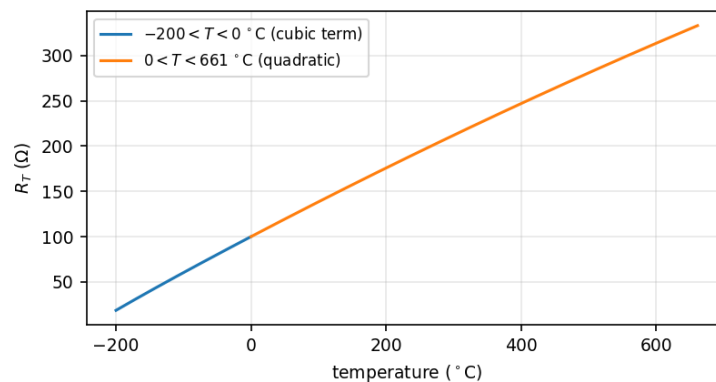


Figure 7: pt100 resistance over its full range; visibly near-linear, in contrast with the thermistor’s exponential.

Worked Example 9.1 (pt100 at two temperatures)

At $T = 100^\circ\text{C}$: $R = 100[1+0.39083-5.775\times 10^{-7}\times 10^4] = 100[1+0.39083-0.005775] = 138.51\ \Omega$.
 At $T = -50^\circ\text{C}$ (cubic branch): $AT = -0.19542$; $BT^2 = -0.0014438$; $(T-100)CT^3 = (-150)(-4.183\times 10^{-12})(-1.25\times 10^5) = -7.84\times 10^{-5}$; $R = 100(1 - 0.19542 - 0.0014438 - 0.0000784) = 80.31\ \Omega$.

10 Sensitivity

Writing the sensor as a transfer function $S_{\text{out}} = f(S_{\text{in}})$, the **sensitivity** is its local slope:

$$S = \frac{\partial S_{\text{out}}}{\partial S_{\text{in}}} \quad \text{e.g.} \quad S = \frac{\Delta \text{current}}{\Delta \text{glucose}}. \quad (17)$$

For a nonlinear sensor, sensitivity depends on the operating point: the lecture's glucose-sensor curve is steep at low concentration and saturates at high concentration, so the same sensor is sensitive at 5 mM and nearly blind at 25 mM.

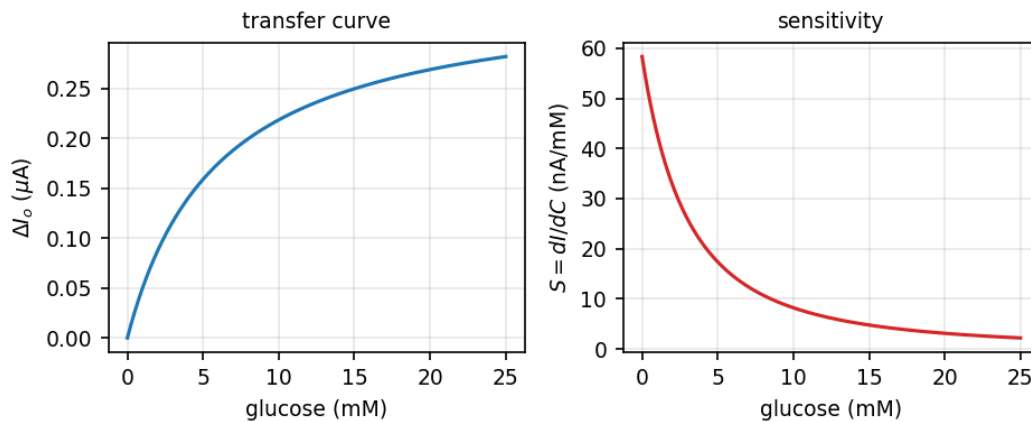


Figure 8: A saturating transfer curve (left) and its derivative, the sensitivity (right): high at low concentration, collapsing as the sensor saturates.

Worked Example 10.1 (Sensitivity of a saturating sensor)

Model the transfer curve as $I(C) = I_{\text{max}}C/(K_m + C)$ with $I_{\text{max}} = 0.35\ \mu\text{A}$, $K_m = 6\ \text{mM}$. Then $S(C) = dI/dC = I_{\text{max}}K_m/(K_m + C)^2$. At $C = 2\ \text{mM}$: $S = 0.35 \times 6/8^2 = 32.8\ \text{nA/mM}$. At $C = 20\ \text{mM}$: $S = 2.1/26^2 = 3.1\ \text{nA/mM}$ — a $10\times$ loss. Design consequence: choose (or dilute to) an operating range on the steep part of the curve.

11 Noise

Noise is random fluctuation of the sensor output essentially unrelated to the input: temperature fluctuations, electromagnetic interference, instability of the sensitive element or electronics, mechanical vibration, fluid-flow artifacts. Quantify it as the root mean square of a sample $(x_1, \dots, x_n)$ of output deviations over a time window:

$$x_{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}}. \quad (18)$$

For zero-mean noise, averaging N independent samples reduces the standard deviation of the *mean* by $1/\sqrt{N}$ — the cheapest noise-reduction tool available in firmware, and the reason the lecture's Arduino sketch streams thousands of samples.

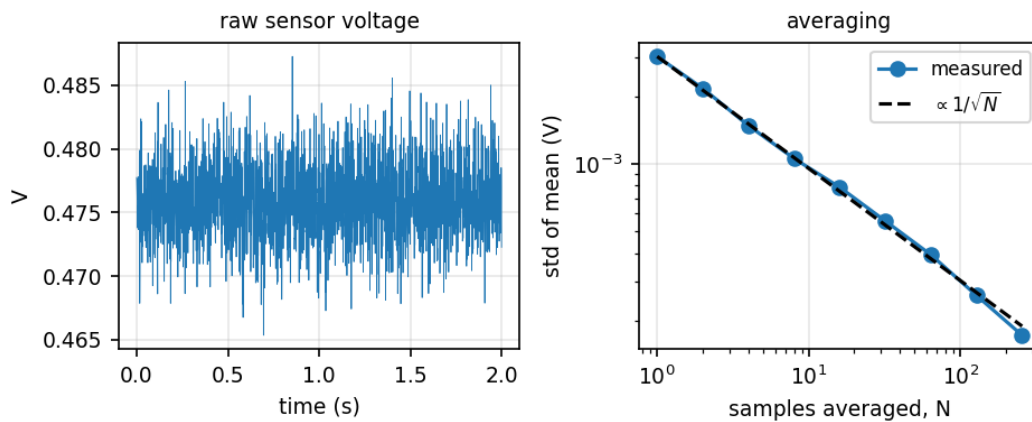


Figure 9: Left: a noisy constant sensor voltage. Right: the standard deviation of an N -sample average falls as $1/\sqrt{N}$.

Worked Example 11.1 (RMS noise from samples)

Five deviations from the mean, in mV: $\{+2, -1, +3, -2, -2\}$.

$$x_{\text{rms}} = \sqrt{\frac{4+1+9+4+4}{5}} = \sqrt{4.4} = 2.10 \text{ mV}.$$

To push the uncertainty of a reported *average* below 0.25 mV requires $N \geq (2.10/0.25)^2 \approx 71$ samples.

PART III

Chemical and Light-Based Sensors (Lecture 16 “Other Sensors”; Notes §§29–30)

12 Chemical Sensors

The notes survey three chemical-sensing families that reappear on the “Other sensors” slide of Lecture 16: smoke, gas, and humidity/moisture.

12.1 Smoke detectors

Two dominant architectures:

- **Ionizing.** A tiny americium radioisotope source ionizes air in an open chamber, letting a small steady current flow from the source’s metal casing to a detector plate. Entering smoke particles — even microscopic ones — attach to the ions and neutralize them, *reducing* the current; the control electronics detect the drop.
- **Optical (photoelectric).** A focused infrared LED and a photodiode sit *off-axis* from one another, so in clean air almost no light reaches the photodiode. Smoke particles scatter LED light toward the photodiode, *increasing* its signal.

Note the complementary logic: ionizing detectors alarm on a current *decrease*, optical detectors on a light *increase*. Both are threshold (switch-like) sensors in the taxonomy of Section 2.

12.2 Gas sensors

Catalytic gas detectors such as the MQ-4 (methane, sensitive down to ~ 200 ppm) contain a small heating element plus a catalytic sensing element whose resistance depends on gas concentration. Usage pattern: supply the heater (often 5 V at a few tens of milliamps), and place the sensing element in a **voltage divider** with a fixed load resistor — exactly the topology of Section 7, with gas concentration playing the role of temperature. Different catalysts give sensitivity to different gases.

Worked Example 12.1 (Gas-sensor divider design)

An MQ-4 sensing element has $R_s \approx 20\text{ k}\Omega$ in clean air, falling to $\approx 5\text{ k}\Omega$ at high methane concentration. With a $10\text{ k}\Omega$ load R_L on a 5 V supply (V_{out} across R_L , sensor on top):

$$V_{\text{out}}^{\text{clean}} = 5 \times \frac{10}{10 + 20} = 1.67\text{ V}, \quad V_{\text{out}}^{\text{gas}} = 5 \times \frac{10}{10 + 5} = 3.33\text{ V}.$$

The $1.67 \rightarrow 3.33\text{ V}$ swing spans half the ADC range — a comfortable design. Choosing $R_L \approx \sqrt{R_s^{\min} R_s^{\max}} = 10\text{ k}\Omega$ (the geometric mean) maximizes the output swing, the same logic as choosing $R_1 = R_0$ for the thermistor.

12.3 Humidity sensors

Where 1–2% accuracy suffices (most applications), low-cost **capacitive** sensors dominate: humidity changes the dielectric constant of a polymer film between two plates, so capacitance tracks relative humidity. Commercial parts integrate the sensing element with a temperature sensor, conditioning electronics, and a serial interface, and achieve their accuracy through **laser calibration** at manufacture — calibration parameters are written into on-chip digital memory (the factory-calibration strategy of Section 5). Unless a specialized application demands extreme accuracy, an IC humidity sensor is the right choice.

13 Light-Based Sensors (Orientation)

Light detection spans optical data transmission, proximity detection, smoke scattering (above), and the photodiode front ends used heavily in the pulse-oximetry lectures. The three passive/junction devices to know:

- **Photoresistor (LDR):** resistance falls with illumination; read with a voltage divider exactly as in Section 7.
- **Photodiode:** generates a small current proportional to incident light; read with a transimpedance amplifier (Part V).
- **Phototransistor:** a photodiode with built-in current gain; simpler to read, slower, less linear.

PART IV

Optical Biosensing and Pulse Oximetry (Lecture 14)

14 Wearables and Vital Signs (Context)

Wearables integrate communication, health, and fitness tracking; the core vital signs are heart rate, blood pressure, breathing rate, and temperature, with combined metrics most useful for preventive care. A bioelectronic measurement system decomposes into: sensors → amplifiers and filters → analog-to-digital converter → signal processing and transmission → display and analytics. Pulse oximetry is the lecture’s worked case study of this whole chain.

15 The Beer–Lambert Law

Light of incident intensity I_{inc} passing through an absorbing medium of path length ℓ emerges attenuated. Each thin slice dx absorbs a fraction of the light proportional to the local intensity and the concentration c of the absorber, which integrates to exponential decay. In the *decadic* convention:

$$I = I_{\text{inc}} 10^{-\varepsilon c \ell}, \quad T \equiv \frac{I}{I_{\text{inc}}} = 10^{-\varepsilon c \ell}, \quad A \equiv -\log_{10} T = \varepsilon c \ell. \quad (19)$$

Definitions and units: **absorbance** A is dimensionless; **transmittance** $T \in [0, 1]$; ε is the molar absorptivity ($\text{M}^{-1}\text{cm}^{-1}$); c the molar concentration; ℓ the path length. Crucially, absorbances of a mixture *add*: $A_{\text{total}} = \sum_i \varepsilon_i c_i \ell$ (same ℓ) — this additivity is what lets pulse oximetry separate two hemoglobin species with two wavelengths.

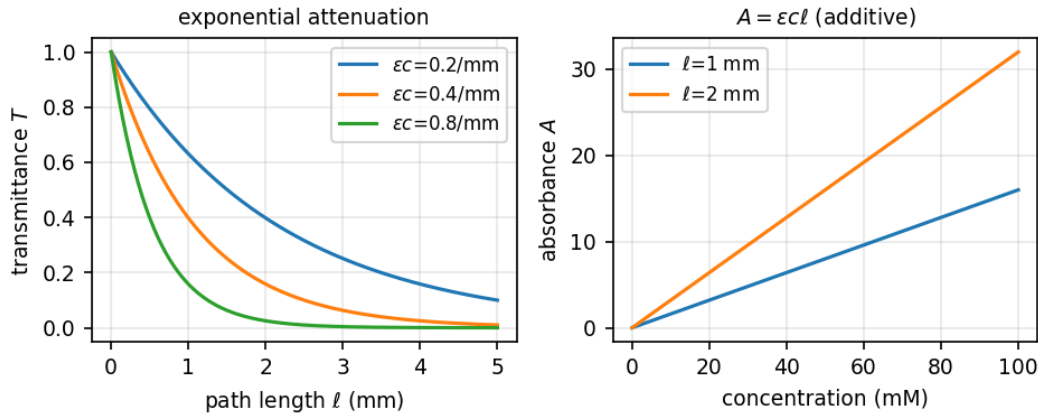


Figure 10: Left: transmittance decays exponentially with path length. Right: absorbance is linear in concentration and path length.

Worked Example 15.1 (Lecture Example 1, in full)

Light at 254 nm is attenuated to 16% of its incident value through a 50 mM benzene solution in a 1.0 mm cuvette. Find A , ϵ , and the transmittance through 2.0 mm.

$$T = 0.16 \Rightarrow A = -\log_{10}(0.16) = 0.796 \approx 0.8.$$

$$\epsilon = \frac{A}{c\ell} = \frac{0.8}{(0.050 \text{ M})(0.1 \text{ cm})} = 160 \text{ M}^{-1}\text{cm}^{-1}.$$

Doubling the path doubles the absorbance: $A_{2\text{mm}} = 1.6$, so $T_{2\text{mm}} = 10^{-1.6} = 0.025 = 2.5\%$. Note the key behavior: absorbance is *additive* in path, transmittance is *multiplicative* ($0.16^2 \approx 0.026$).

16 Pulse Oximetry

16.1 Principle

Pulse oximetry measures blood oxygenation by spectrophotometry at two wavelengths, exploiting the different extinction spectra of oxygenated (HbO_2) and deoxygenated (Hb) hemoglobin:

$$\text{SO}_2 = \frac{C_{\text{HbO}_2}}{C_{\text{HbO}_2} + C_{\text{Hb}}}. \quad (20)$$

At **red** (≈ 660 nm) Hb absorbs much more than HbO_2 ; at **near-infrared** (≈ 880 – 940 nm) the ordering reverses. Two LEDs alternate flashes through (or into) perfused tissue, and one photodetector reads both. Hypoxemia — lower than normal blood oxygenation — shows up as a changed ratio between the two channels.

16.2 The photoplethysmogram: AC and DC

The detected light at each wavelength has:

- a large **DC** component — absorption by skin pigmentation, bone, venous blood, and the non-pulsating part of arterial blood;
- a small **AC** component — the extra absorption from arterial blood pulsing with the cardiac cycle (path length swings between diastole $d_{\min}$ and systole $d_{\max}$; detected intensity swings between I_H and I_L).

The AC-over-DC waveform is the **photoplethysmogram (PPG)**. Normalizing each channel by its own DC removes LED brightness, detector gain, and tissue differences:

$$R_{OS} = \frac{AC_{\text{red}}/DC_{\text{red}}}{AC_{\text{IR}}/DC_{\text{IR}}} \quad (21)$$

An empirical **calibration curve** maps $R_{OS} \rightarrow \text{SpO}_2$; the Beer–Lambert theoretical curve is close but deviates because of scattering, so manufacturers fit a second-order polynomial from human calibration data. A widely used linear approximation over the clinical range is $\text{SpO}_2 \approx 110 - 25 R_{OS}$ (so $R_{OS} \approx 0.5$ corresponds to $\text{SpO}_2 \approx 97.5\%$).

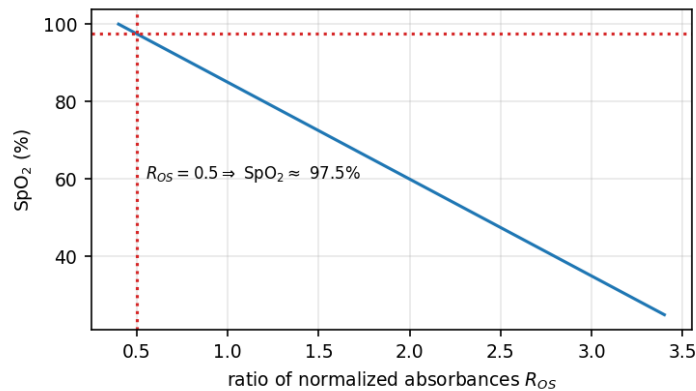


Figure 11: Linearized empirical calibration: $\text{SpO}_2 \approx 110 - 25R_{OS}$.

Worked Example 16.1 (Lecture ratio example)

Measured PPG amplitudes: red AC = 20 mV on DC = 625 mV; infrared AC = 20 mV on DC = 330 mV.

$$R_{OS} = \frac{20/625}{20/330} = \frac{0.0320}{0.0606} = 0.528.$$

With the linear calibration, $\text{SpO}_2 \approx 110 - 25(0.528) = 96.8\%$ — a healthy saturation. Note that the raw AC amplitudes were *equal*; only normalization by the DC levels reveals the physiology.

16.3 Transmission vs. reflectance geometry

Transmission mode (LEDs and photodetector on opposite sides) works only on tissue thin enough to transilluminate — fingertips and earlobes. **Reflectance mode** (LEDs and detector side by side) frees the sensor to sit anywhere; measured AC amplitude is highest at the forehead, mid-range on the arms, and weak on the legs and chest, so the forehead is the best reflectance site.

17 An Integrated AFE: the MAX30102

The MAX30102 is an integrated pulse-oximetry/heart-rate module — LEDs (660 nm red, 880 nm IR), photodiode, optics, and low-noise electronics with ambient-light rejection — i.e., a complete **analog front end (AFE)** presenting a digital I²C interface. Key facts to reason with quantitatively:

- Supplies: 1.8 V core, separate 3.3 V for LEDs (V_{LED+} with bypass capacitors); five wires to the microcontroller: power, ground, SCL (clock), SDA (data), INT (active-low interrupt).
- An 18-bit current-input ADC digitizes the photodiode; ambient light cancellation and a digital filter precede a 32-sample FIFO data register.
- Sample rate is programmable: 50, 100, 200, 400, 800, 1000, 1600, or 3200 samples/s; each SpO₂ “sample” is one red + one IR conversion (3 bytes per channel in the FIFO). The sample rate caps the usable LED pulse width.
- Die-temperature readings (for wavelength compensation) need only be taken about once a second.

Worked Example 17.1 (FIFO servicing deadline)

At 50 samples/s the 32-deep FIFO fills in $32/50 = 0.64$ s, so the host must service the interrupt at least every 640 ms or data is lost. At 1000 sps the deadline shrinks to 32 ms. This is why the part has an interrupt line rather than relying on polling.

Worked Example 17.2 (Choosing a sample rate)

The PPG’s useful content lies below ~ 10 Hz (fundamental 0.7–3.5 Hz plus a handful of harmonics). Nyquist requires $f_s > 2 \times 10 = 20$ Hz; the minimum setting of 50 sps already gives a $2.5\times$ margin, and higher rates mainly buy averaging (noise reduction $\propto 1/\sqrt{N}$, Section 11) at the cost of LED power.

PART V

Interface Electronics: Op-Amps (Lecture 16)

18 Differential Measurement

Place two nominally identical sensing elements side by side: R_1 exposed to the analyte (*sensing*), R_2 shielded from it (*not sensing*, the reference). Both experience the same temperature drift, supply wander, and ambient interference; only R_1 sees the analyte. The difference $S = R_1 - R_2$ cancels the common-mode disturbances and leaves the analyte response — visibly cleaner than either raw channel. The Wheatstone bridge (Section 8) is exactly this idea in resistor form, and the op-amp's differential input (below) is its amplifier form.

19 The Operational Amplifier

An op-amp amplifies the difference of its inputs with enormous gain:

$$V_{\text{out}} = A(V_+ - V_-), \quad A \sim 10^3\text{--}10^9. \quad (22)$$

The three **Golden Rules** for circuits with negative feedback:

1. No current flows into the inputs V_+ and V_- .
2. The input voltages are equal: $V_+ = V_-$ (the “virtual short”).
3. The output can drive whatever current is required.

Rule 2 is a consequence of the huge gain: a few-volt output divided by $A \geq 10^5$ forces the input difference below tens of microvolts. With the bare (open-loop) amplifier, even a 0.5 V input difference would demand an output of $10^3 \times 0.5 = 500$ V — the output simply saturates at the supply rails; useful linear circuits therefore always employ feedback.

19.1 Inverting amplifier

Input through R_i to the $-$ node; feedback R_f from output to the same node; $+$ grounded. By Rule 2 the $-$ node sits at 0 V (virtual ground); by Rule 1 the two resistor currents are equal:

$$\frac{V_{\text{in}} - 0}{R_i} = \frac{0 - V_{\text{out}}}{R_f} \implies \boxed{V_{\text{out}} = -\frac{R_f}{R_i} V_{\text{in}}} \quad (23)$$

Worked Example 19.1 (Lecture example)

$R_i = 10\text{ k}\Omega$, $R_f = 100\text{ k}\Omega$: $V_{\text{out}} = -(100/10)V_{\text{in}} = -10 V_{\text{in}}$. A 50 mV sensor signal becomes -0.5 V.

19.2 Non-inverting amplifier

Input drives V_+ directly; R_i runs from the $-$ node to ground and R_f from the output back to the $-$ node. Rule 2 sets $V_- = V_{in}$; Rule 1 equates the currents through R_i and R_f :

$$\frac{0 - V_{in}}{R_i} = \frac{V_{in} - V_{out}}{R_f} \implies \boxed{V_{out} = V_{in} \frac{R_i + R_f}{R_i} = \left(1 + \frac{R_f}{R_i}\right) V_{in}} \quad (24)$$

With the same resistors as above the gain is $+11$ (not $+10$): the non-inverting gain can never fall below 1.

19.3 Transimpedance (photodiode readout)

A photodiode outputs a *current* I_s proportional to light. Feed it into the $-$ node (virtual ground) with feedback R_f ; all of I_s must flow through R_f (Rule 1):

$$\frac{0 - V_{out}}{R_f} = I_s \implies \boxed{V_{out} = -I_s R_f} \quad (25)$$

The virtual ground also holds the photodiode at zero bias, which improves its linearity — the reason this circuit, not a plain resistor, is the standard photodiode front end (it is Stage 1 of the lecture's oximeter read circuit; Stage 2 measures and subtracts the DC before further amplifying the AC, so the small PPG signal can use the full ADC range).

Worked Example 19.2 (Lecture design problem)

The photodiode delivers $I_s = 10 \mu\text{A}$ at full signal and the ADC wants $|V_{out}| = 3 \text{ V}$. Then

$$R_f = \frac{3 \text{ V}}{10 \mu\text{A}} = 300 \text{ k}\Omega, \quad V_{out} = -3 \text{ V}.$$

If ambient light doubles I_s , the output saturates — hence the MAX30102's ambient-light cancellation and the two-stage AC/DC separation.

Worked Example 19.3 (Bridge + amplifier budget)

The strain bridge of Section 8 delivers 1.65 mV full scale. To span a 3.3 V ADC we need gain $3.3/0.00165 = 2000$. A single inverting stage with $R_i = 1 \text{ k}\Omega$ would need $R_f = 2 \text{ M}\Omega$; in practice this is split (e.g., two stages of $\sim 45\times$) or done with an instrumentation amplifier, but the budget arithmetic is exactly this.

PART VI

Signal Processing for Sensor Data (Lectures 14 and 16)

20 Analog vs. Digital, Sampling

An analog signal is continuous in time and value (4.9372...); a digital signal is a sequence of quantized samples ([10 100 1000...]). Sampling at rate f_s represents content only below the Nyquist frequency $f_s/2$; quantization adds the $\pm\frac{1}{2}$ LSB error of Section 3. Everything below assumes uniformly sampled data.

21 Time Domain to Frequency Domain

The **Fourier transform** decomposes a signal into constituent frequencies and their amplitudes: a pure sine occupies a single spectral line; a square wave requires an infinite series of odd harmonics with amplitudes falling as $1/k$:

$$x_{sq}(t) = \frac{4}{\pi} \sum_{k \text{ odd}} \frac{\sin(2\pi k f t)}{k} = \frac{4}{\pi} \left[\sin(2\pi f t) + \frac{1}{3} \sin(6\pi f t) + \frac{1}{5} \sin(10\pi f t) + \dots \right]. \quad (26)$$

Partial sums show the mechanics: one harmonic is a sine; adding $3f$, $5f$, ... sharpens the edges (Figure 12). In discrete time the **FFT** computes this decomposition; selecting spectral components and inverse-transforming reconstructs the signal — *one* component gives a pure sine at the dominant rate, a few components give a caricature (visible “signal loss”), all components return the original exactly.

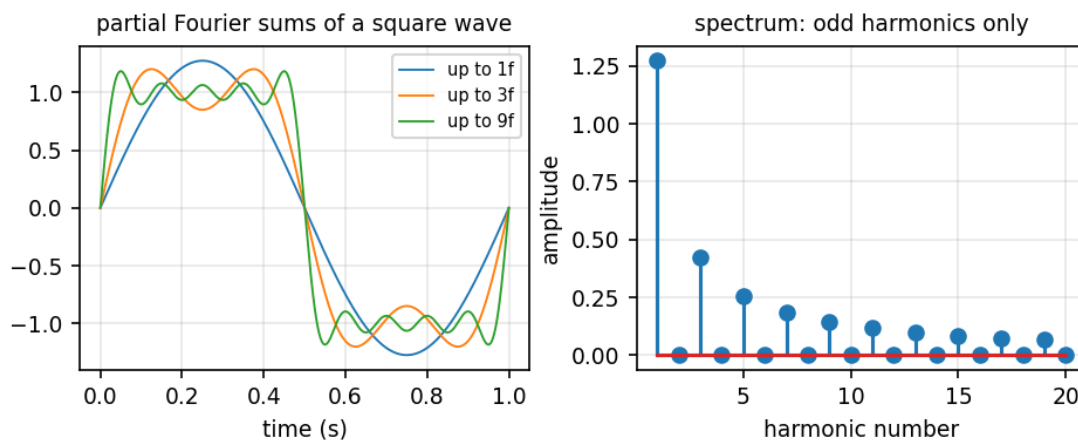


Figure 12: Partial Fourier sums of a square wave (left) and its odd-harmonic spectrum (right).

22 Filters

A filter passes some frequency bands and attenuates others. The four canonical types, each specified by cutoff/corner frequencies where the response is -3 dB (half power):

- **Low-pass:** passes below f_C ; removes high-frequency noise (“smoothing”).
- **High-pass:** passes above f_C ; removes DC and slow drift.
- **Band-pass:** passes f_1 – f_2 ; isolates a signal of known rhythm (e.g., PPG at 0.5–5 Hz).
- **Notch (band-stop):** removes a narrow band (e.g., 50/60 Hz mains pickup).

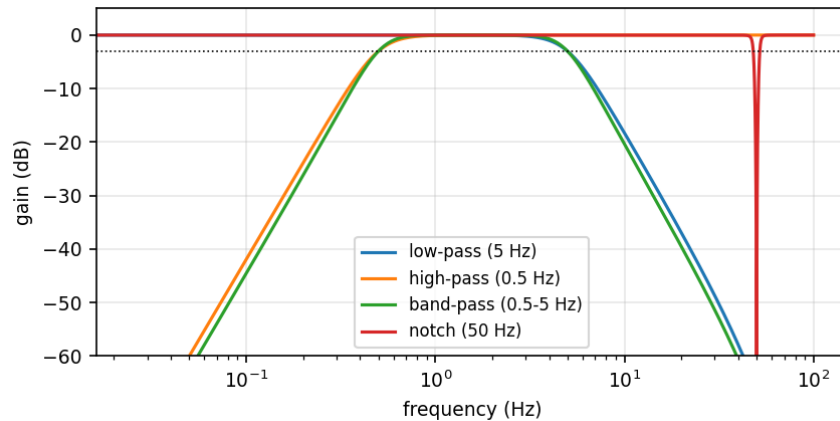


Figure 13: Frequency responses of the four filter types (Butterworth designs; dotted line marks -3 dB).

Worked Example 22.1 (Choosing a PPG filter)

Heart rate spans roughly 40–210 BPM = 0.67–3.5 Hz; useful PPG harmonics extend to ~ 10 Hz; motion/baseline drift lives below ~ 0.5 Hz; mains pickup at 50/60 Hz. A band-pass of about 0.5–10 Hz (or low-pass at 10 Hz plus baseline subtraction) keeps the physiology and rejects both drift and mains — exactly the smoothing + drift-correction pipeline below.

23 The PPG Processing Pipeline

The lecture’s end-to-end recipe, applied to a synthetic 72-BPM PPG with drift and noise (all figures generated by the listing below):

1. **Smoothing** (moving average = crude low-pass): removes high-frequency noise; the FFT after smoothing shows the noise floor collapsed away from the signal peaks.
2. **Drift correction:** estimate the baseline with a much longer moving average (or detrending) and subtract it, centering the waveform on zero — a crude high-pass.
3. **Peak detection:** locate systolic peaks (and valleys) with a minimum spacing and prominence constraint.
4. **Heart rate:** from consecutive peak times t_i , each beat-to-beat interval gives $\text{HR}_i = 60/(t_{i+1} - t_i)$ BPM; alternatively, the FFT’s dominant peak f_{pk} in the physiologic band gives $\text{HR} = 60f_{\text{pk}}$.

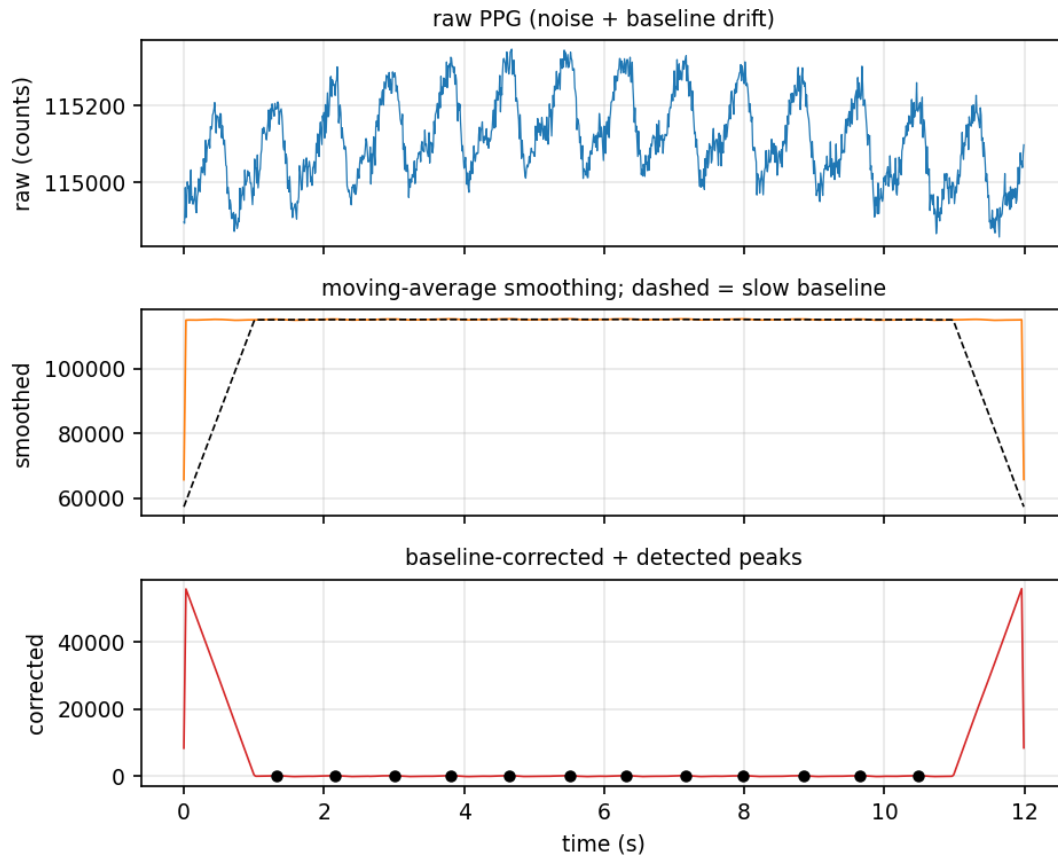


Figure 14: Raw synthetic PPG, smoothed trace with estimated baseline, and baseline-corrected signal with detected peaks.

Worked Example 23.1 (Heart rate from peak times)

Detected peaks at $t = 1.30, 2.10, 2.96, 3.80, 4.60$ s give intervals 0.80, 0.86, 0.84, 0.80 s and beat rates 75.0, 69.8, 71.4, 75.0 BPM — mean over the full record 72.0 BPM (the beat-to-beat spread comes from noise perturbing individual peak locations). The FFT of the same record peaks at 1.20 Hz = 72.0 BPM. Interval-based HR responds beat by beat (useful for heart rate *variability*); FFT-based HR averages over the window and is more noise-robust.

Listing 4: Complete PPG pipeline: smooth, detrend, find peaks, estimate HR two ways.

```
import numpy as np
from scipy.signal import find_peaks

fs = 100.0                                # samples per second
t = np.arange(0, 12, 1/fs)
f0 = 1.2                                  # 72 BPM ground truth
pulse = (0.6*np.sin(2*np.pi*f0*t - np.pi/2)
         + 0.25*np.sin(2*np.pi*2*f0*t + 0.6)
         + 0.10*np.sin(2*np.pi*3*f0*t + 1.1))
drift = 0.9*np.sin(2*np.pi*0.05*t) + 0.05*t
rng = np.random.default_rng(1)
```

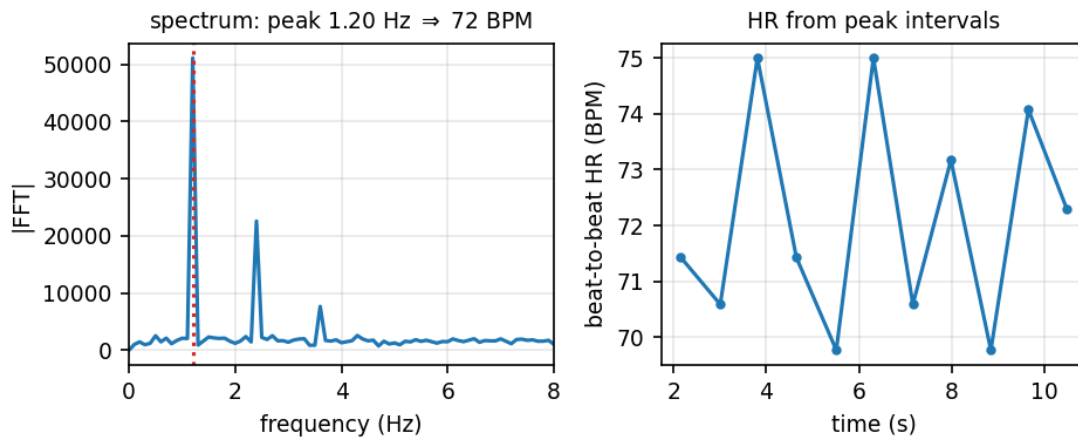


Figure 15: Spectrum of the corrected PPG (peak at 1.20 Hz ⇒ 72 BPM) and beat-to-beat heart rate from peak intervals.

```
raw    = 115000 + 200*pulse + 150*drift + 25*rng.standard_normal(t.size)

movavg  = lambda x, w: np.convolve(x, np.ones(w)/w, mode="same")
smooth  = movavg(raw, 7)                                # 1. smoothing (low-pass)
base    = movavg(smooth, int(2*fs) | 1)                  # 2. slow baseline estimate
corr    = smooth - base                                  # drift-corrected PPG

edge = int(1.0*fs)                                       # discard filter edge artifacts
pk, _ = find_peaks(corr[edge:-edge], distance=int(0.5*fs), prominence=80)
pk += edge                                              # 3. peak detection

hr_beats = 60.0 / np.diff(t[pk])                        # 4a. beat-to-beat HR
seg = corr[edge:-edge] - corr[edge:-edge].mean()
X, f = np.fft.rfft(seg), np.fft.rfftfreq(seg.size, 1/fs)
band = (f > 0.5) & (f < 4.0)                            # physiologic band only
hr_fft = 60.0 * f[band][np.argmax(np.abs(X)[band])]     # 4b. spectral HR
print(hr_beats.mean(), hr_fft)                         # -> 72.1, 72.0 BPM
```

Two implementation details worth internalizing from the listing: moving averages produce *edge artifacts* (the window runs off the data), so trim the edges before peak-finding and FFTs; and restrict the spectral peak search to the physiologic band, or residual drift energy at very low frequency will masquerade as an absurd heart rate.

A Errata in the Source Notes

1. **Section 27.3:** “raw value straight from a 12-bit ADC is a number between 0 and 1023” — 0–1023 is *10-bit*; 12-bit spans 0–4095.
2. **Examples 28.1/28.3:** exponent written with 25 + 237; should be 25 + 273.15. Corrected result for Example 28.3: 1.14 V, not 1.15 V.
3. **Notation:** “°K” should be K; “principal of calibration” should be “principle.”
4. **Dangling references:** several “see Figure ” citations in the notes have empty numbers.

B Formula Sheet

ADC step	$\Delta V = V_{\text{ref}}/2^N$; error $\pm \frac{1}{2}$ LSB
Linear interpolation	$y = y_1 + (y_2 - y_1) \frac{x - x_1}{x_2 - x_1}$
β model	$1/T = 1/T_0 + (1/\beta) \ln(R/R_0)$; $R = R_0 e^{\beta(1/T - 1/T_0)}$
Steinhart–Hart	$1/T = A + B \ln R + C(\ln R)^3$
Divider	$V_{\text{out}} = V_{\text{in}} R_1 / (R_1 + R)$; $R = R_1 (V_{\text{in}} / V_{\text{out}} - 1)$; slope $V_{\text{in}} \beta / 4T_0^2$ at $R=R_1$
Wheatstone	$V_o = V_s \frac{R_1 R_4 - R_2 R_3}{(R_1 + R_2)(R_3 + R_4)} \approx V_s \frac{\delta R}{4R}$ (equal arms)
RTD (pt100, $T > 0$)	$R_T = R_0[1 + AT + BT^2]$, $A=3.9083\text{e-}3$, $B=-5.775\text{e-}7$
Sensitivity	$S = \partial S_{\text{out}} / \partial S_{\text{in}}$
RMS noise	$x_{\text{rms}} = \sqrt{\frac{1}{n} \sum x_i^2}$; averaging: $\sigma_{\bar{x}} = \sigma / \sqrt{N}$
Self-heating	$\Delta T = P / \delta$; divider: $P = V_{\text{in}}^2 R / (R_1 + R)^2$
Beer–Lambert	$T = 10^{-\varepsilon c \ell}$; $A = -\log_{10} T = \varepsilon c \ell$; mixtures add
Oximetry ratio	$R_{\text{OS}} = \frac{\text{AC}_{\text{red}} / \text{DC}_{\text{red}}}{\text{AC}_{\text{IR}} / \text{DC}_{\text{IR}}}$; $\text{SpO}_2 \approx 110 - 25 R_{\text{OS}}$
Op-amp	$V_{\text{out}} = A(V_+ - V_-)$; rules: no input current, $V_+ = V_-$, output drives any load
Inverting / non-inv.	$-R_f / R_i$ / $1 + R_f / R_i$
Transimpedance	$V_{\text{out}} = -I_s R_f$
Square wave	$\frac{4}{\pi} \sum_{k \text{ odd}} \sin(2\pi k f t) / k$
Heart rate	$\text{HR} = 60 / \Delta t_{\text{peak}}$ BPM $= 60 f_{\text{pk}}$
